(T) Roll No. .....

PAPER ID—21135

B.Tech. EXAMINATION, 2025

(Fourth Semester)

COMPUTER SCIENCE AND ENGINEERING

Programming in Java

Time : 3 Hours *Maximum Marks : 70*

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Section. Q. No. 1 is compulsory.

(Compulsory Question)

1. (a) What is the relation between JVM, JRE and JDK ? 2

(b) Differentiate between static and dynamic binding. 2

(c) Show usage of 'super' keyword with a suitable example. 2

(d) Explain the role of 'finally' keyword. 2

(e) Differentiate between Swing and AWT frameworks. 2

(f) How list in Java is different from normal array ? Explain with example. 2

(g) Explain 'javap' tool with an example. 2

Section I

2. (a) What are the key mechanisms employed by the Java Virtual Machine (JVM) to ensure both platform independence and security and how do these mechanisms contribute to Java's suitability for building secure and portable applications ?

(b) Illustrate the compilation and execution process of Java program with the help of a flow chart. 7+7

3. (a) How does the security architecture of the Java Virtual Machine (JVM) interact with its security policy to establish a robust security framework ? Discuss the implications of this architecture in ensuring the integrity and confidentiality of Java applications.

(b) Discuss the various data types available in Java. Explain size of each data type. What type casting options are available among them ? 7+7

Section II

4. Create a Java program to calculate the area of shapes using method overloading. Define a "ShapeCalculator" class with overloaded methods for computing the area of circles, rectangles and triangles. Each method should accept parameters relevant to the shape being calculated. Ensure error handling for invalid inputs and provide a user interface for inputting parameters and displaying results. Test the program with different inputs for accuracy and reliability. 14

5. Develop a Java program to simulate a university system using inheritance. Create a superclass "Person" with attributes like "name" and "age." Implement subclasses for different roles within the university, such as "Student," "Professor," and "Staff," inheriting from the Person superclass. Each subclass should have specific attributes and methods, such as "studentID" for students and "department" for professors. Additionally, include methods for displaying information and performing role-specific actions, such as enrolling in courses for students and grading assignments for professors. Test the program by creating instances of each role and demonstrating inheritance and method overriding. 14

Section III

6. (a) Compare creating threads by extending the Thread class versus implementing the Runnable interface in Java. Analyze the advantages and disadvantages of each approach, considering factors like code reusability and resource management.

(b) Discuss the importance of layout managers for arranging components within containers. Compare BorderLayout, FlowLayout, GridLayout, and CardLayout, highlighting their characteristics and use cases. 7+7

7. (a) Evaluate the necessity of organizing classes into packages in Java for better code organization, maintenance and reusability. Discuss the role of access modifiers (public, protected, private and default) in controlling class visibility within and across packages.

(b) Develop a Java program that prompts the user for input and validates it using custom exceptions. Define custom exception classes to handle specific error conditions (e.g., invalid input format, out-of-range values). Implement try-catch blocks to catch these custom exceptions and provide appropriate error messages to the user, guiding them to correct their input. 7+7

Section IV

8. Write short notes on the following. Write example programs also. 7+7

(a) List Iterator

(b) Hash Maps.

9. (a) Discuss I/O streams in Java. Elaborate on stream filters and buffered streams.

(b) Illustrate how PL/SQL stored procedures are created with an example. 7+7